

Capture the Flag 4: The Student List

Goal: Build a scrollable student directory using Dart Lists and `ListView.builder`, replacing manually duplicated widgets with dynamically generated UI.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- Explain an error (not fix your error)
- Clarify documentation

You may NOT ask AI to generate the solution/code for a flag.

Flag 0 — Create Your Feature Branch

Objective: Create a feature branch before building your student list.

Create:

`feature/student-list`

Screenshot:

- Terminal showing your current branch

```
MINGW64:/c/Users/satom/OneDrive - TAFE NSW/IT314AB_Verano

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/data-ui)
$ git status
On branch feature/data-ui
Your branch is up to date with 'origin/feature/data-ui'.

nothing to commit, working tree clean

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/data-ui)
$ git branch
* feature/data-ui
  feature/widgets
  main

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/data-ui)
$ git checkout -b feature/student-list
Switched to a new branch 'feature/student-list'

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-list)
$ git branch
  feature/data-ui
* feature/student-list
  feature/widgets
  main
```

🚩 Flag 1 — Build Your Student Data

Objective: Store multiple students inside a Dart `List`.

Research what a Dart `List` is and why it is useful when handling multiple pieces of similar data.

Instead of creating one student at a time with separate variables, create a collection that stores at least five students.

Each student should contain information such as:

- Image
- Name
- Course
- Year Level
- Age
- Hobby

Your data should exist separately from your UI widgets.

Screenshot:

Code:

- Your student list
- At least five student entries

```
13 class Profile {
14     String image;
15     String name;
16     String course;
17     String yearLevel;
18     int age;
19     String hobby;
20
21     Profile({
22         required this.image,
23         required this.name,
24         required this.course,
25         required this.yearLevel,
26         required this.age,
27         required this.hobby,
28     });
29 }
30
31 List<Profile> profiles = [
32     Profile(
33         image: 'assets/kierstien.jpeg',
34         name: 'Verano, Kierstien S.',
35         course: 'BSIT',
36         yearLevel: '3rd Year',
37         age: 20,
38         hobby: 'Sleeping',
39     ),
40
41     Profile(
42         image: 'assets/kyla.jpeg',
43         name: 'Caballero, Kyla Marie S.',
44         course: 'BSIT',
45         yearLevel: '3rd Year',
46         age: 20,
47         hobby: 'Drawing',
48     ),
49
```

```
50     Profile(
51         image: 'assets/zelon.jpeg',
52         name: 'Estimo, Zelon Matthew C.',
53         course: 'BSIT',
54         yearLevel: '3rd Year',
55         age: 21,
56         hobby: 'Dancing',
57     ),
58
59     Profile(
60         image: 'assets/leila.jpeg',
61         name: 'Bangoy, Leila G.',
62         course: 'BSIT',
63         yearLevel: '3rd Year',
64         age: 20,
65         hobby: 'Watching Movies',
66     ),
67
68     Profile(
69         image: 'assets/cassy.jpeg',
70         name: 'Oraiz, Cassandra Gayle R.',
71         course: 'BSIT',
72         yearLevel: '3rd Year',
73         age: 20,
74         hobby: 'Gaming',
75     ),
76 ];
```

Flag 2 — The Copy-Paste Trap

You already know how to create a Student Card.

You created five student cards manually.

You duplicated the same card design until all five students appear.

Stop.

Imagine having:

- 50 students.
- 500 students.
- 5,000 students.

Would copying the same widget over and over still be practical?

This flag is intentionally making you feel the problem before solving it.

Screenshot:

- Your five manually created student cards (do not change your past code yet)

```
142     children: [
143       Card(
144         color: Colors.white.withOpacity(0.3),
145         elevation: 0,
146         child: Padding(
147           padding: const EdgeInsets.all(8),
148           child: Column(
149             mainAxisAlignment: MainAxisAlignment.center,
150             children: [
151               SizedBox(height: 50),
152               Image.asset('assets/kierstien.jpeg', width: 130),
153               SizedBox(height: 30),
154               Text(
155                 'Verano, Kierstien S.',
156                 style: TextStyle(
157                   fontSize: 20,
158                   fontWeight: FontWeight.bold,
159                 ), // TextStyle
160               ), // Text
161               SizedBox(height: 10),
162               Text('BSIT', style: TextStyle(fontSize: 18)),
163               SizedBox(height: 10),
164               Text('3rd Year', style: TextStyle(fontSize: 18)),
165               SizedBox(height: 10),
166               Text('Age: 20', style: TextStyle(fontSize: 18)),
167               SizedBox(height: 10),
168               Text('Hobby: Sleeping', style: TextStyle(fontSize: 18)),
169               SizedBox(height: 50),
170             ],
171           ), // Column
172         ), // Padding
173       ), // Card
174     ],
```

```
187     Card(
188       color: Colors.white.withOpacity(0.3),
189       elevation: 0,
190       child: Padding(
191         padding: const EdgeInsets.all(8),
192         child: Column(
193           mainAxisAlignment: MainAxisAlignment.center,
194           children: [
195             SizedBox(height: 50),
196             Image.asset('assets/kyla.jpeg', width: 130),
197             SizedBox(height: 30),
198             Text(
199               'Caballero, Kyla Marie S.',
200               style: TextStyle(
201                 fontSize: 20,
202                 fontWeight: FontWeight.bold,
203               ), // TextStyle
204             ), // Text
205             SizedBox(height: 10),
206             Text('BSIT', style: TextStyle(fontSize: 18)),
207             SizedBox(height: 10),
208             Text('3rd Year', style: TextStyle(fontSize: 18)),
209             SizedBox(height: 10),
210             Text('Age: 20', style: TextStyle(fontSize: 18)),
211             SizedBox(height: 10),
212             Text('Hobby: Drawing', style: TextStyle(fontSize: 18)),
213             SizedBox(height: 50),
214           ],
215         ), // Column
216       ), // Padding
217     ), // Card
218   ],
```

```

231 Card(
232   color: Colors.white.withOpacity(0.3),
233   elevation: 0,
234   child: Padding(
235     padding: const EdgeInsets.all(8),
236     child: Column(
237       mainAxisAlignment: MainAxisAlignment.center,
238       children: [
239         SizedBox(height: 50),
240
241         Image.asset('assets/zelon.jpeg', width: 130),
242
243         SizedBox(height: 30),
244
245         Text(
246           'Estimo, Zelon Matthew C.',
247           style: TextStyle(
248             fontSize: 20,
249             fontWeight: FontWeight.bold,
250           ), // TextStyle
251         ), // Text
252
253         SizedBox(height: 10),
254
255         Text('BSIT', style: TextStyle(fontSize: 18)),
256
257         SizedBox(height: 10),
258
259         Text('3rd Year', style: TextStyle(fontSize: 18)),
260
261         SizedBox(height: 10),
262
263         Text('Age: 21', style: TextStyle(fontSize: 18)),
264
265         SizedBox(height: 10),
266
267         Text('Hobby: Dancing', style: TextStyle(fontSize: 18)),
268
269         SizedBox(height: 50),
270       ],
271     ), // Column
272   ), // Padding
273 ), // Card
274

```

```

275 Card(
276   color: Colors.white.withOpacity(0.3),
277   elevation: 0,
278   child: Padding(
279     padding: const EdgeInsets.all(8),
280     child: Column(
281       mainAxisAlignment: MainAxisAlignment.center,
282       children: [
283         SizedBox(height: 50),
284
285         Image.asset('assets/leila.jpeg', width: 130),
286
287         SizedBox(height: 30),
288
289         Text(
290           'Bangoy, Leila G.',
291           style: TextStyle(
292             fontSize: 20,
293             fontWeight: FontWeight.bold,
294           ), // TextStyle
295         ), // Text
296
297         SizedBox(height: 10),
298
299         Text('BSIT', style: TextStyle(fontSize: 18)),
300
301         SizedBox(height: 10),
302
303         Text('3rd Year', style: TextStyle(fontSize: 18)),
304
305         SizedBox(height: 10),
306
307         Text('Age: 20', style: TextStyle(fontSize: 18)),
308
309         SizedBox(height: 10),
310
311         Text(
312           'Hobby: Watching Movies',
313           style: TextStyle(fontSize: 18),
314         ), // Text
315
316         SizedBox(height: 50),
317       ],
318     ), // Column
319   ), // Padding
320 ), // Card
321

```

```

322 Card(
323   color: Colors.white.withOpacity(0.3),
324   elevation: 0,
325   child: Padding(
326     padding: const EdgeInsets.all(8),
327     child: Column(
328       mainAxisAlignment: MainAxisAlignment.center,
329       children: [
330         SizedBox(height: 50),
331
332         Image.asset('assets/cassy.jpeg', width: 130),
333
334         SizedBox(height: 30),
335
336         Text(
337           'Oraiz, Cassandra Gayle R.',
338           style: TextStyle(
339             fontSize: 20,
340             fontWeight: FontWeight.bold,
341           ), // TextStyle
342         ), // Text
343
344         SizedBox(height: 10),
345
346         Text('BSIT', style: TextStyle(fontSize: 18)),
347
348         SizedBox(height: 10),
349
350         Text('3rd Year', style: TextStyle(fontSize: 18)),
351
352         SizedBox(height: 10),
353
354         Text('Age: 20', style: TextStyle(fontSize: 18)),
355
356         SizedBox(height: 10),
357
358         Text('Hobby: Gaming', style: TextStyle(fontSize: 18)),
359
360         SizedBox(height: 50),
361       ],
362     ), // Column
363   ), // Padding
364 ), // Card

```

🚩 Flag 3 — Discover **ListView**

Objective: Make your student directory scrollable using **ListView**.

Research Flutter's **ListView**.

Replace your previous layout with a scrolling layout so users can browse through all of your student cards naturally.

Your application should now allow continuous vertical scrolling.

Test

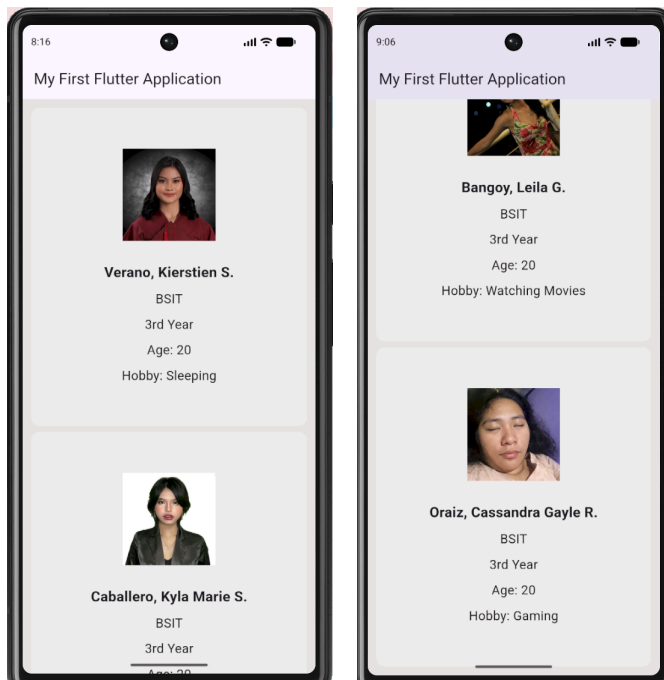
The user should be able to:

- Scroll from the first student
- Continue scrolling
- Reach the last student without resizing the screen

Screenshot:

- **ListView** in your code
- Top of the application
- Bottom portion showing later students

```
135 body: ListView(  
136 children: [
```



🚩 Flag 4 — Unlock `ListView.builder`

Objective: Generate student cards automatically using `ListView.builder`.

Research `ListView.builder`.

Instead of writing five separate Student Cards, generate them automatically from your student list.

Your UI should now create one card for each student in your list.

Test

Add one new student to your list.

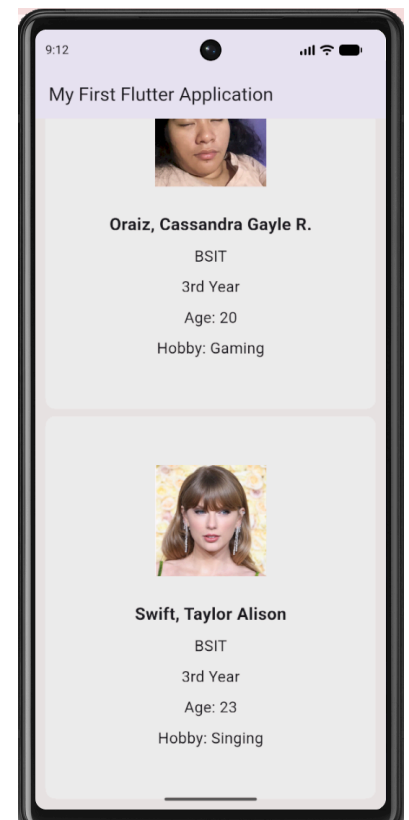
Without creating another card manually, your application should display the new student automatically.

Screenshot:

- `ListView.builder` in your code
- Student list
- Newly generated student appearing in the app

```
123     body: ListView.builder(  
124       padding: const EdgeInsets.all(8),  
125       itemCount: profiles.length,  
126       itemBuilder: (context, index) {  
127         return profileCard(profiles[index]);  
128       },  
129     ), // ListView.builder
```

```
31 List<Profile> profiles = [  
32   Profile(  
33     image: 'assets/kierstien.jpeg',  
34     name: 'Verano, Kierstien S.',  
35     course: 'BSIT',  
36     yearLevel: '3rd Year',  
37     age: 20,  
38     hobby: 'Sleeping',  
39   ),  
40  
41   Profile(  
42     image: 'assets/kyla.jpeg',  
43     name: 'Caballero, Kyla Marie S.',  
44     course: 'BSIT',  
45     yearLevel: '3rd Year',  
46     age: 20,  
47     hobby: 'Drawing',  
48   ),  
49  
50   Profile(  
51     image: 'assets/zelon.jpeg',  
52     name: 'Estimo, Zelon Matthew C.',  
53     course: 'BSIT',  
54     yearLevel: '3rd Year',  
55     age: 21,  
56     hobby: 'Dancing',  
57   ),  
58 ]  
  
59 Profile(  
60   image: 'assets/leila.jpeg',  
61   name: 'Bangoy, Leila G.',  
62   course: 'BSIT',  
63   yearLevel: '3rd Year',  
64   age: 20,  
65   hobby: 'Watching Movies',  
66 ),  
67  
68 Profile(  
69   image: 'assets/cassy.jpeg',  
70   name: 'Oraiz, Cassandra Gayle R.',  
71   course: 'BSIT',  
72   yearLevel: '3rd Year',  
73   age: 20,  
74   hobby: 'Gaming',  
75 ),  
76  
77 Profile(  
78   image: 'assets/taylor.webp',  
79   name: 'Swift, Taylor Alison',  
80   course: 'BSIT',  
81   yearLevel: '3rd Year',  
82   age: 23,  
83   hobby: 'Singing',  
84 ),  
85 ];  
86
```



🚩 Flag 5 — Every Student Deserves an ID

Objective: Expand your student records with additional information.

Real student directories contain more than just names.

Add at least three new fields to every student.

Examples:

- Student ID
- Email
- Favorite Subject

Update your Student Card so the new information appears consistently across every generated card.

Screenshot:

- Updated student data
- Student Card displaying the additional fields

```
13  class Profile {
14      String image;
15      String name;
16      String course;
17      String yearLevel;
18      int age;
19      String hobby;
20      String studentId;
21      String email;
22      String favoriteSubject;
23
24      Profile({
25          required this.image,
26          required this.name,
27          required this.course,
28          required this.yearLevel,
29          required this.age,
30          required this.hobby,
31          required this.studentId,
32          required this.email,
33          required this.favoriteSubject,
34      });
35  }
```

```
37  List<Profile> profiles = [
38      Profile(
39          image: 'assets/kierstien.jpeg',
40          name: 'Verano, Kierstien S.',
41          course: 'BSIT',
42          yearLevel: '3rd Year',
43          age: 20,
44          hobby: 'Sleeping',
45          studentId: '11111',
46          email: 'kierstien.verano@dbtc-cebu.edu.ph',
47          favoriteSubject: 'Break Time',
48      ),
49
50      Profile(
51          image: 'assets/kyla.jpeg',
52          name: 'Caballero, Kyla Marie S.',
53          course: 'BSIT',
54          yearLevel: '3rd Year',
55          age: 20,
56          hobby: 'Drawing',
57          studentId: '22222',
58          email: 'kyla.caballero@dbtc-cebu.edu.ph',
59          favoriteSubject: 'Break Time',
60      ),
61
62      Profile(
63          image: 'assets/zelon.jpeg',
64          name: 'Estimo, Zelon Matthew C.',
65          course: 'BSIT',
66          yearLevel: '3rd Year',
67          age: 21,
68          hobby: 'Dancing',
69          studentId: '33333',
70          email: 'zelon.estimo@dbtc-cebu.edu.ph',
71          favoriteSubject: 'Dismissal Time',
72      ),
73  ]
```



```

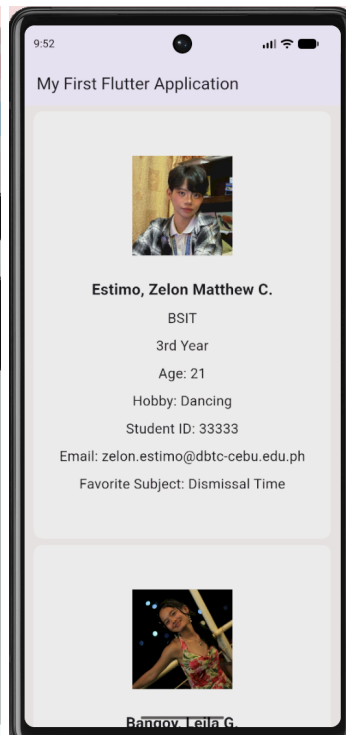
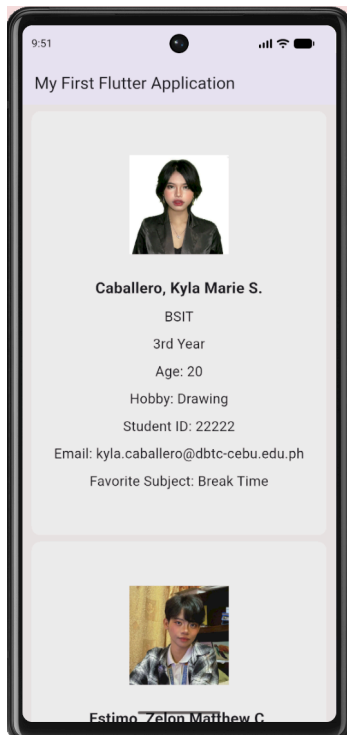
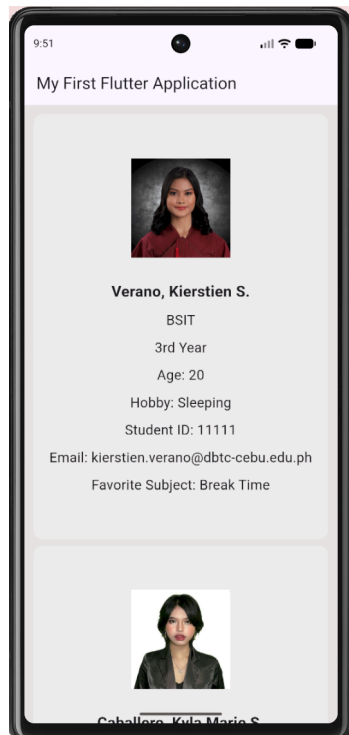
74 Profile(
75   image: 'assets/leila.jpeg',
76   name: 'Bangoy, Leila G.',
77   course: 'BSIT',
78   yearLevel: '3rd Year',
79   age: 20,
80   hobby: 'Watching Movies',
81   studentId: '44444',
82   email: 'leila.bangoy@dbtc-cebu.edu.ph',
83   favoriteSubject: 'Application Development and Emerging Technologies',
84 ),
85
86 Profile(
87   image: 'assets/cassy.jpeg',
88   name: 'Oraiz, Cassandra Gayle R.',
89   course: 'BSIT',
90   yearLevel: '3rd Year',
91   age: 20,
92   hobby: 'Gaming',
93   studentId: '55555',
94   email: 'cassandra.oraiz@dbtc-cebu.edu.ph',
95   favoriteSubject: 'Internet of Things',
96 ),
97
98 Profile(
99   image: 'assets/taylor.webp',
100  name: 'Swift, Taylor Alison',
101  course: 'BSIT',
102  yearLevel: '3rd Year',
103  age: 23,
104  hobby: 'Singing',
105  studentId: '66666',
106  email: 'taylor.swift@dbtc-cebu.edu.ph',
107  favoriteSubject: 'Fundamentals of Databased System',
108 ),
109 ];
110

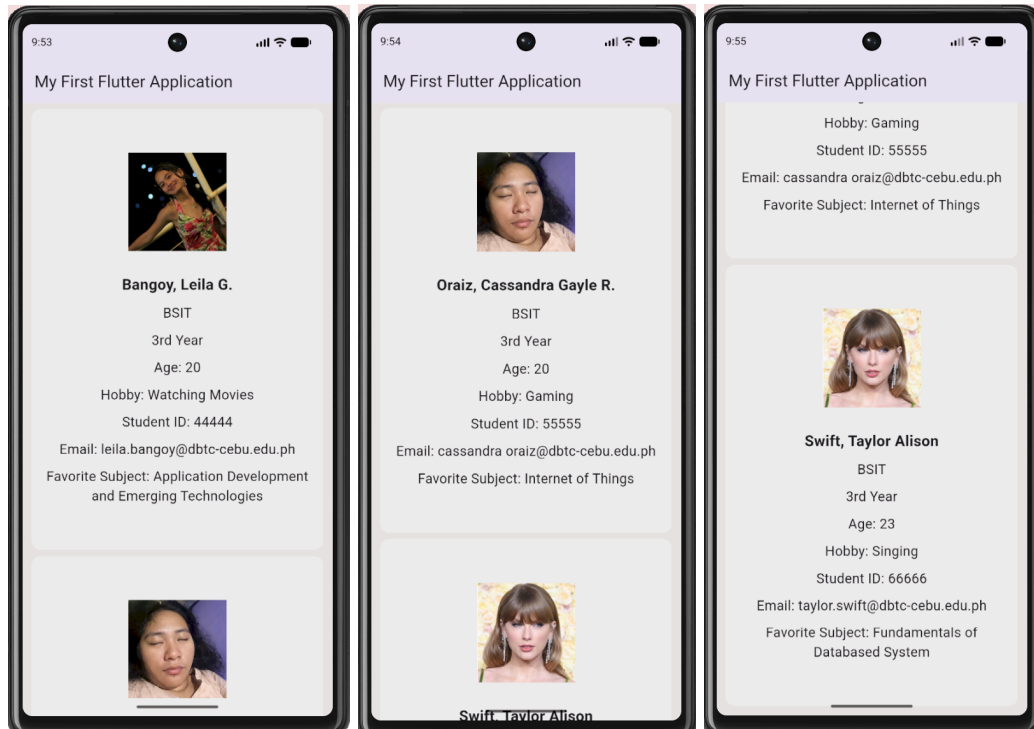
```

```

160
161 Text(
162   "Student ID: ${profile.studentId}",
163   textAlign: TextAlign.center,
164   style: TextStyle(fontSize: 18),
165 ), // Text
166 SizedBox(height: 10),
167 Text(
168   "Email: ${profile.email}",
169   textAlign: TextAlign.center,
170   style: TextStyle(fontSize: 18),
171 ), // Text
172 SizedBox(height: 10),
173 Text(
174   "Favorite Subject: ${profile.favoriteSubject}",
175   textAlign: TextAlign.center,
176   style: TextStyle(fontSize: 18),
177 ), // Text

```





🚩 Flag 6 — Empty Classroom

What happens when there are no students?

Objective: Handle an empty **List** gracefully.

Imagine your app just connected to a database.

The database returns:

[]

No students.

What should the user see?

Instead of displaying a broken or confusing screen, research how Flutter can display a different widget when a condition is met.

Create an empty state that clearly communicates that there are currently no students available.

Examples:

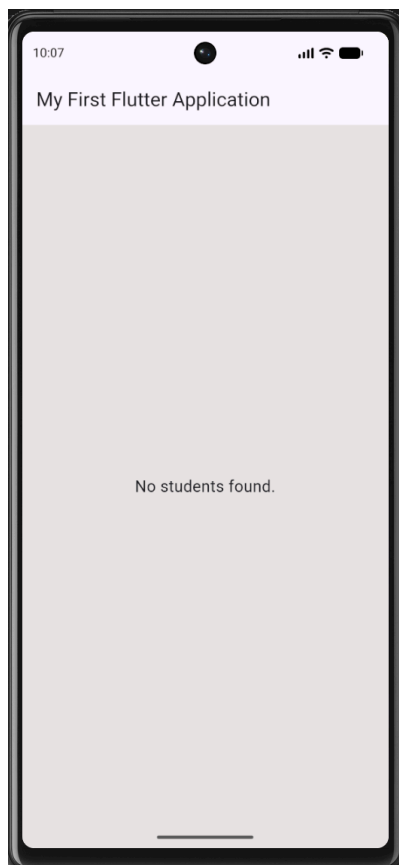
- "No students found."
- "Student list is empty."

Test

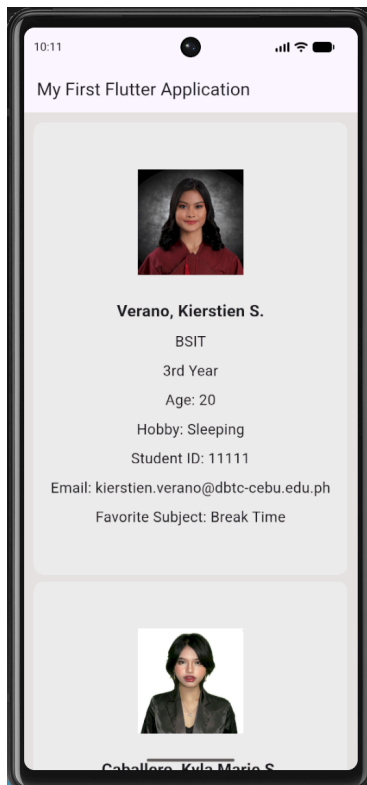
- Display your student list normally.
- Then temporarily make your list empty.
- Your application should display the empty state instead of student cards.

Screenshot:

- Empty student list
- Empty-state UI
- Application still functioning correctly



```
37 List<Profile> profiles = [];
```



🚩 Flag 7 — Sort the Roll Call

Objective: Change the order of your student list.

Class lists are rarely displayed randomly.

Research how Dart can rearrange items inside a `List`.

Sort your students alphabetically by name.

Test

Before sorting:

- Maria
- John
- Aaron
- Sophia

After sorting:

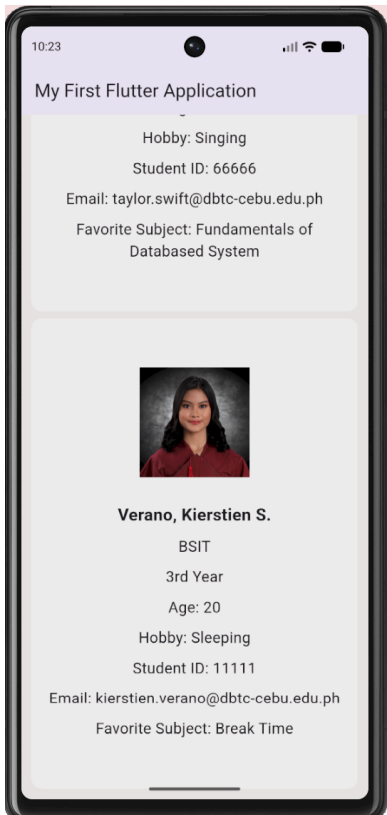
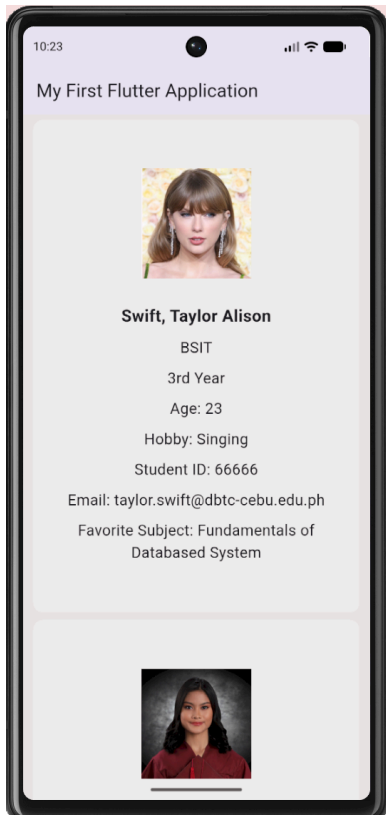
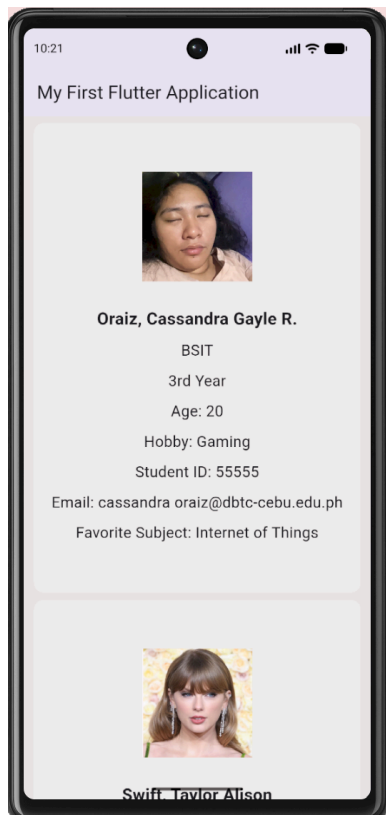
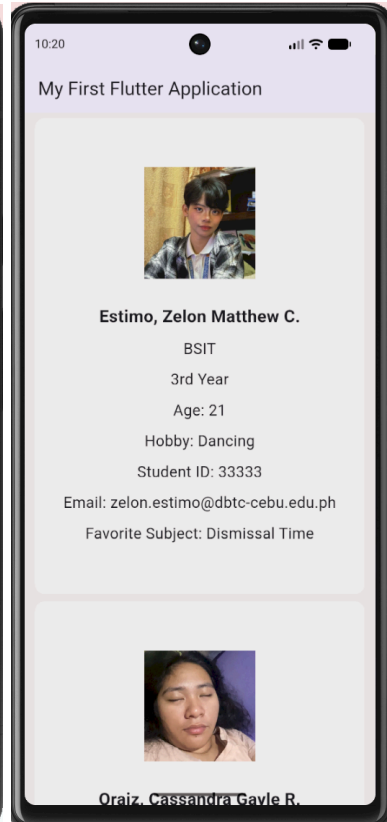
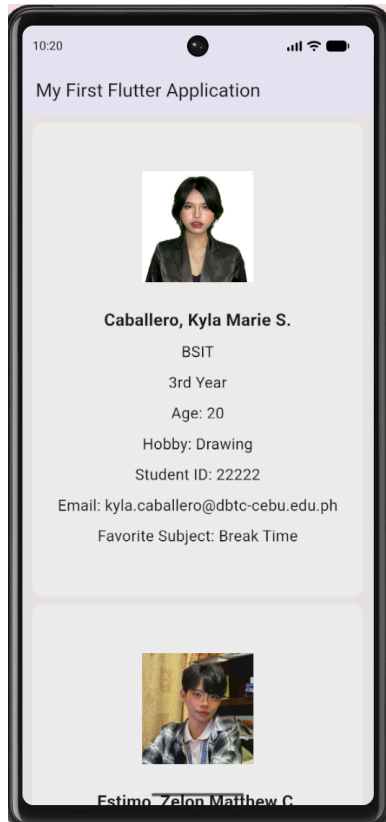
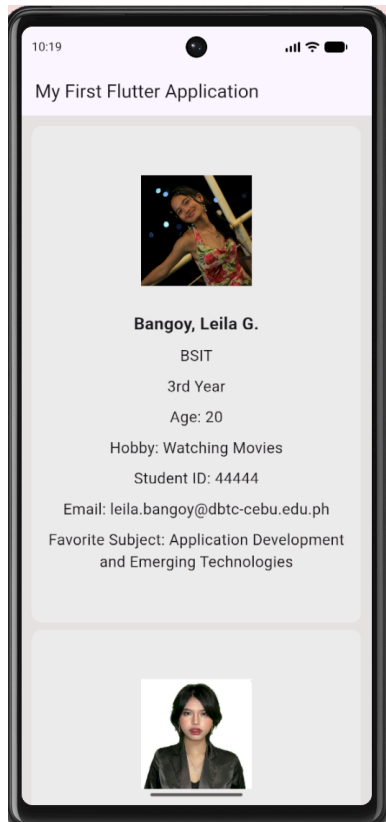
- Aaron
- John
- Maria
- Sophia

Your UI should immediately reflect the new order.

Screenshot:

- Student list before or after sorting
- Application displaying the sorted order

```
Run | Debug | Profile
111 void main() {
112   profiles.sort((a, b) => a.name.compareTo(b.name));
113   // ma'am ako gi sort lang by surname instead sa first name since sa akong profile ga una ang mga surnames. thank you
114   runApp(const MyApp());
115 }
```



🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Student List CTF.

Use an appropriate commit message following proper Git standards.

Branch:

feature/student-list

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
MINGW64:/c:/Users/satom/OneDrive - TAFE NSW/IT314AB_Verano

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-list)
$ git branch
  feature/data-ui
* feature/student-list
  feature/widgets
  main

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-list)
$ git status
On branch feature/student-list
Your branch is up to date with 'origin/feature/student-list'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   FlutterProjects/my_first_flutter_app/lib/main.dart
    modified:   FlutterProjects/my_first_flutter_app/pubspec.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    FlutterProjects/my_first_flutter_app/assets/taylor.webp

no changes added to commit (use "git add" and/or "git commit -a")

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-list)
$ git add .

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-list)
$ git commit -m "feat: submit The Student List"
[feature/student-list 463c15c] feat: submit The Student List
3 files changed, 120 insertions(+), 52 deletions(-)
 create mode 100644 FlutterProjects/my_first_flutter_app/assets/taylor.webp

satom@Tomilings MINGW64 ~/OneDrive - TAFE NSW/IT314AB_Verano (feature/student-list)
$ git push
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 20 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 82.64 KiB | 20.66 MiB/s, done.
Total 9 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/KVerano/IT314AB_Verano.git
   f9861d2..463c15c  feature/student-list -> feature/student-list
```